Utkarsh Srivastava

9956971456 | utkarsh.srivastava2023@vitstudent.ac.in | India

EDUCATION

Vellore Institute of Technology Aug. 2023 – Present
Bachelor of Technology in Computer Science and Engineering Vellore, TN

- **CGPA**: 9.1
- Ranked 6^{th} out of 150 in the Computer Science and Engineering (Bioinformatics) Batch

AISSCE Apr. 2021 – Mar. 2022
12^{th} *Grade* Lucknow, UP

- **Score**: 91%
- **Subjects**: Physics, Chemistry, Mathematics, Biology, English

PROJECTS

On Non-Stationary Constrained Multi-Armed Bandits

- Engineered two adaptive variants of the Constraint Lower Confidence Bound algorithm (SW-CON-LCB and D-CON-LCB) using sliding window and exponential discounting mechanisms to manage piecewise-stationary environments.
- Derived theoretical upper bounds for expected suboptimal and infeasible arm selections, proving a regret growth rate of $\mathcal{O}(T^{\frac{1+v}{2}})$ under abrupt distributional shifts with $\mathcal{O}(T^v)$ breakpoints.
- Validated theoretical guarantees through Monte Carlo simulations and implementation on an AMR dataset, demonstrating substantial reductions in cumulative regret compared to stationary baselines.

Adaptive Security Selection Framework for Resource-Constrained IoT Devices Using LinUCB

- Designed a multi-armed bandit model that incorporated cryptographic/security constraints into arm selection and regret minimisation.
- Studied trade-offs between performance, feasibility, and security under constrained sequential decision-making frameworks.

The Impact of Network Topology on the Performance of 6CoCoA in 6TiSCH Networks

- Implemented and evaluated the 6CoCoA congestion control algorithm in Contiki-NG across multiple 25-node and 64-node network topologies under varying traffic loads.
- Analysed delivery ratio and end-to-end delay, showing strong topology-dependent performance with mesh/star networks outperforming linear bottleneck structures.

Computational Biomarker Analysis of Alzheimers Disease

- Processed and analysed the GSE5281 genomic dataset to identify potential computational markers associated with Alzheimers disease progression.
- Applied data preprocessing and computational analysis techniques to extract meaningful biological insights, demonstrating the ability to bridge computer science and bioinformatics for medical research.

ACADEMIC SUBMISSIONS

- Utkarsh Srivastava, Ashish R Hota. “On Non-Stationary Constrained Multi-Armed Bandits.” to be Submitted to TMLR, 2026.
- Utkarsh Srivastava, Avidha Halder, Selvi M. “An Adaptive Security Selection Framework for Resource-Constrained IoT Devices Using the LinUCB Algorithm.” Under Review: INDISCON, 2026.

SKILLS

Languages : Python, C, SQL (SQLPlus), R, Java
Core Interests : Machine Learning, Theoretical Computer Science, Computational Biology
Tools : NumPy, AlphaFold2, Pymol

EXPERIENCE

Culinary Club Jan. 2025 – Present
Chair Person & Co-Secretary Vellore Institute of Technology, Vellore

- Leads planning and execution of large-scale culinary workshops and campus events, coordinating timelines, vendors, and participant outreach.
- Oversees logistics, internal communications, and cross-team collaboration for events with 100+ attendees.
- Manages operational documentation, scheduling, and workflow organisation to ensure smooth delivery of club initiatives.